

Dataset A: Sparse Features (10 relevant out of 1000)

BEST: L1 (Lasso) Regularization

Reasoning:

- Coefficient Distribution: L1 produces sharp sparsity with 954 exact zeros compared to L2's broad distribution.
- Feature Selection: Automatically identifies and retains only around 46 relevant features.
- Optimization Path: The loss surface is non-smooth due to the absolute value penalty, which drives some coefficients exactly to zero, resulting in a sparse model.
- Efficiency: Simplifies the model while maintaining excellent performance ($R^2 \approx 1.0$).

Dataset B: Dense Features (all contribute meaningful information)

BEST: L2 (Ridge) Regularization

Reasoning:

- Coefficient Distribution: Smooth, bell-shaped distribution preserving all 1000 features.
- No Information Loss: Shrinks coefficients proportionally without eliminating any of them.
- Optimization Path: The loss surface is non-smooth due to the absolute value penalty, which drives some coefficients exactly to zero, resulting in a sparse model.
- Performance: Achieves the best R^2 and lowest MSE by retaining all informative features.

Note: L1's aggressive sparsity (around 950 zeros) discards valuable information.

Dataset C: Grouped Correlated Features (5 relevant groups)

BEST: Elastic Net Regularization

Reasoning:

- Group-wise Selection: The average coefficient magnitude shows clear separation between relevant and irrelevant groups.
- Stability: The L2 component handles within-group correlation smoothly; L1 alone is unstable with correlated features.
- Controlled Sparsity: Achieves around 313 non-zero coefficients versus L2's 1000, effectively removing irrelevant groups.
- Optimization Path: The loss surface is non-smooth due to the absolute value penalty, which drives some coefficients exactly to zero, resulting in a sparse model.